

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

- A) Draw the following sentence according to drawing standard SP 46. (or any other standard convention)

Remember

6

GOOD job, INSTAGRAM, Work is Worship.

- B) Draw a regular pentagon of 30 mm side.

Remember

6

- C) Explain different methods of dimensioning by drawing suitable diagrams.

Understand

6

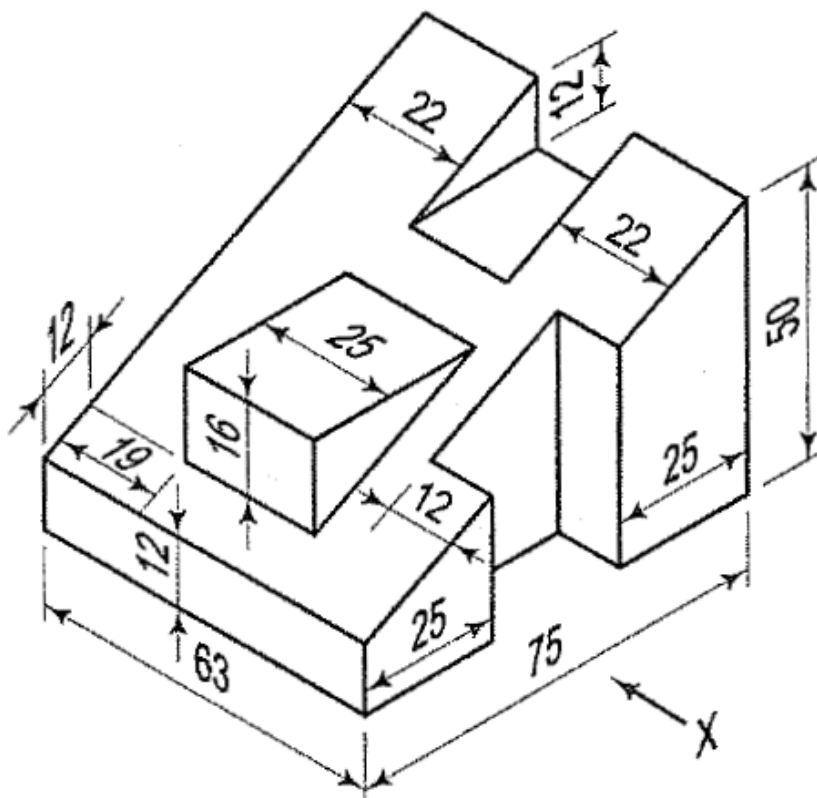
Q.2 Solve Any one of the following.

12

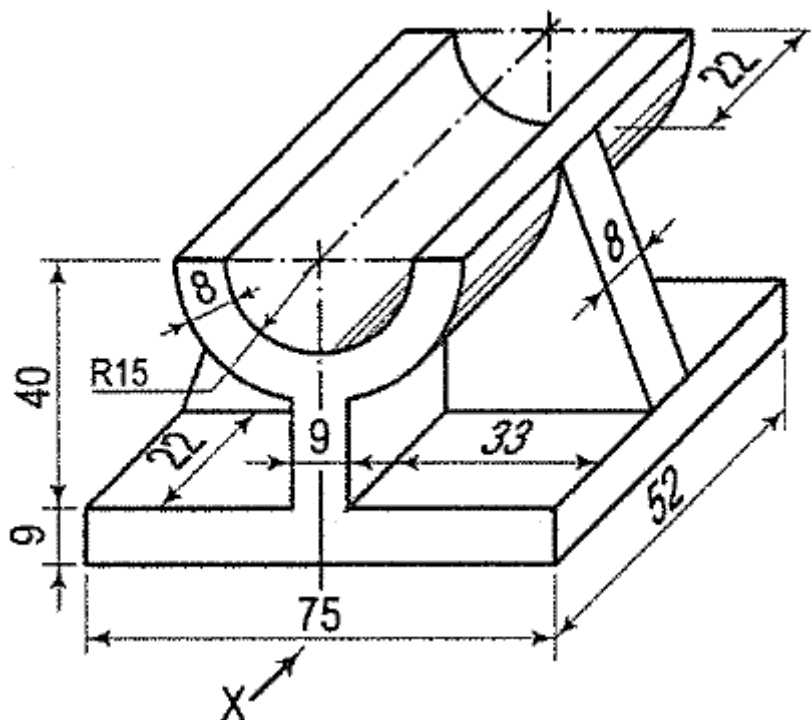
- A) Fig, shows the pictorial view of an object, draw its front view looking from direction of X, TV and left hand side view by using First angle Projection method.

Apply

12



- B) Fig, shows the pictorial view of an object, draw its front view looking from direction of X, TV and right hand side view by using Third angle Projection method. 12



Q. 3 Solve Any Two of the following.

12

- A) A line AB of 70 mm long has its end 'A' 15 mm from both H.P. and V.P. The other end B is 40 mm above H.P. and 60 mm in front of V.P. Draw the projections of the line and determine the inclination of the line with H.P. and V.P. Line is in first quadrant. 6 Evaluate
- B) A square ABCD of 50 mm side has its corner A in the HP, its diagonal AC inclined at 30° to the H.P. and the diagonal BD inclined at 45° to the V.P. and parallel to the H.P. Draw its projection 6 Evaluate
- C) A rectangle length of side 50 mm and 80 mm has its shortest side AB in H.P. and incline 30° to V.P. Complete the projection of plane if surface of the plane is inclined 45° to H.P. 6 Evaluate

Q.4 Solve Any One of the following.

12

- A) A cone of diameter of base 50 mm and height 75 mm is resting on H.P. on the point of its periphery of the base. The axis of the cone is inclined to H.P. by 30° and top view of the axis inclined at 45° with V.P. Draw the projection when apex is nearer to the observer. 12 Evaluate
- B) A pentagonal pyramid base 25 mm side and axis 50 mm long has one of its triangular faces in the V.P. and edge of the base contained by that face makes an angle of 30° with the H.P. Draw its projections. 12 Evaluate

Q. 5 Solve Any one of the following.

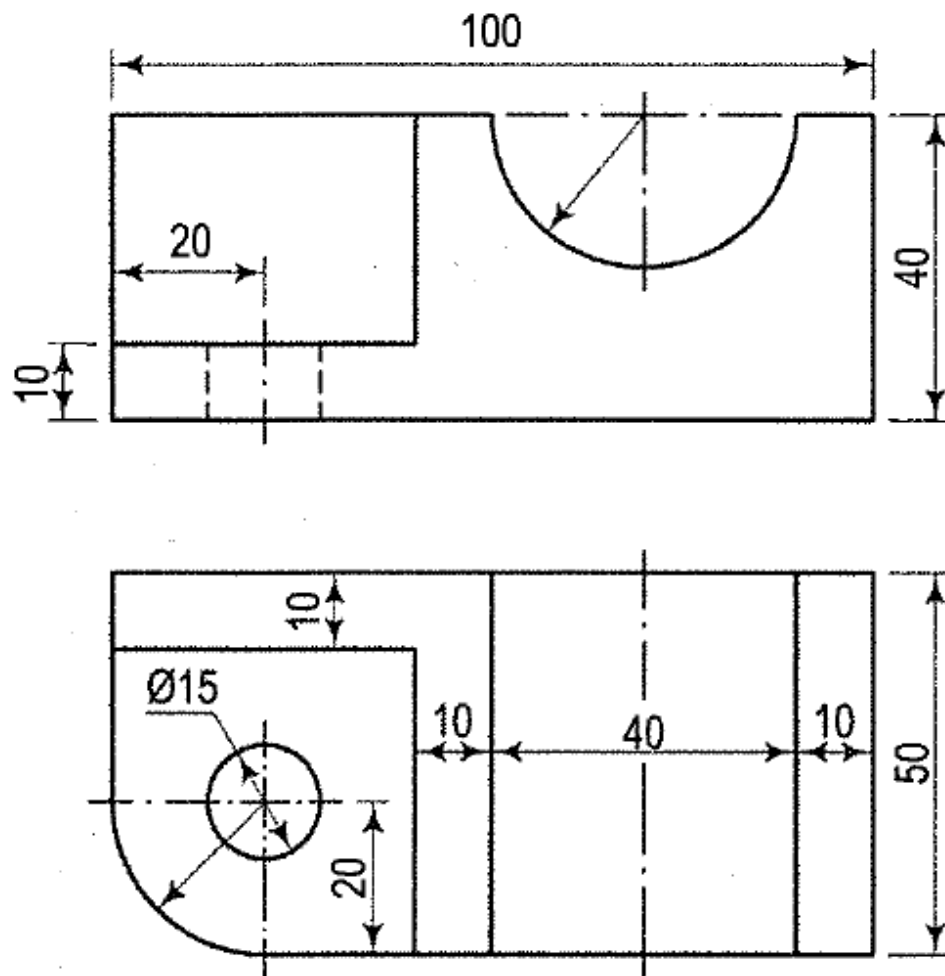
12

- A) A hexagonal pyramid, base 30 mm side and axis 70 mm long is resting on its slant of the face on the horizontal plane. A section plane, perpendicular to the V.P. inclined to the H.P. passes through the highest corner of the base and intersecting the axis at 25 mm from the base. Draw the projections of the solid and determine the inclination of the section plane with the H.P. Evaluate

12

- B) Draw the isometric view of the following object having FV and TV drawn by first angle projection method. Synthesize

12



*** End ***